BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Internet layer.

Q2. Stack.

Q3. Tracking variable state changes as an algorithm runs.

Q4. Eight.

You Do pages · Independent questions

Q1. Hostname lookup [2], address records returned/cached [2], browser uses IP to connect to server [2].

Q2. DNS maps names/records [3], routers forward packets using IP destination/network information [3].

Q3. Resolver caching [2], TTL expiry [2], changed authoritative record [2], client/network diagnostic tests [2].

Exit Ticket · Model answer

False. DNS supplies name-resolution data; web content is retrieved separately.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

LESSON 144 · Lesson 144 · DNS and IP Routing: From Name to Destination

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which TCP/IP layer handles routing and IP addressing?

Q2. What data structure uses LIFO?

Q3. What is a trace table used for?

Q4. How many bits are in one byte?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

DNS — Hierarchical name-resolution system.

IP address — Network-layer address used to route packets.

resolver — Client/service that queries DNS and caches answers.

router — Forwards IP packets towards destination networks.

New learning

Resolve hostname — Browser requests address for a website name through DNS.

Use IP destination — Application connects to a resolved IP address and transport port.

Route across networks — Routers forward packets hop by hop to the destination network.

Worked example

A pupil types example.org and receives an IP address from DNS. Explain what DNS does and does not do.

1. Name example.org — Identify the relevant information and make a prediction.

2. DNS returns an IP address — Explain the change using correct technical terms.

3. Connect to web server — DNS maps the hostname to relevant resource records such as an A/AAAA address. DNS does not directly transmit the web page. Browser makes a separate connection to the chosen server address and requests content, often over HTTPS.

We Do · Practise with guidance

A DNS answer has been cached but website server has moved. Explain a possible issue and resolution.

Cached old address

Identify the relevant information and make a prediction.

Server address changes

Explain the change using correct technical terms.

Refresh after TTL

The cached address may remain until its TTL expires; client may contact an outdated endpoint. The resolver can refresh after expiry. DNS propagation and authoritative changes do not guarantee immediate worldwide consistency.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain how DNS helps a user retrieve a website by name. [6]

Question 2

Differentiate DNS name resolution from IP packet routing. [6]

Question 3

A school website moves servers but some users reach the old address. Explain cached DNS and propose sensible checks. [8]

Exit Ticket · Answer before checking

DNS sends the entire web page to the browser after it receives the hostname.